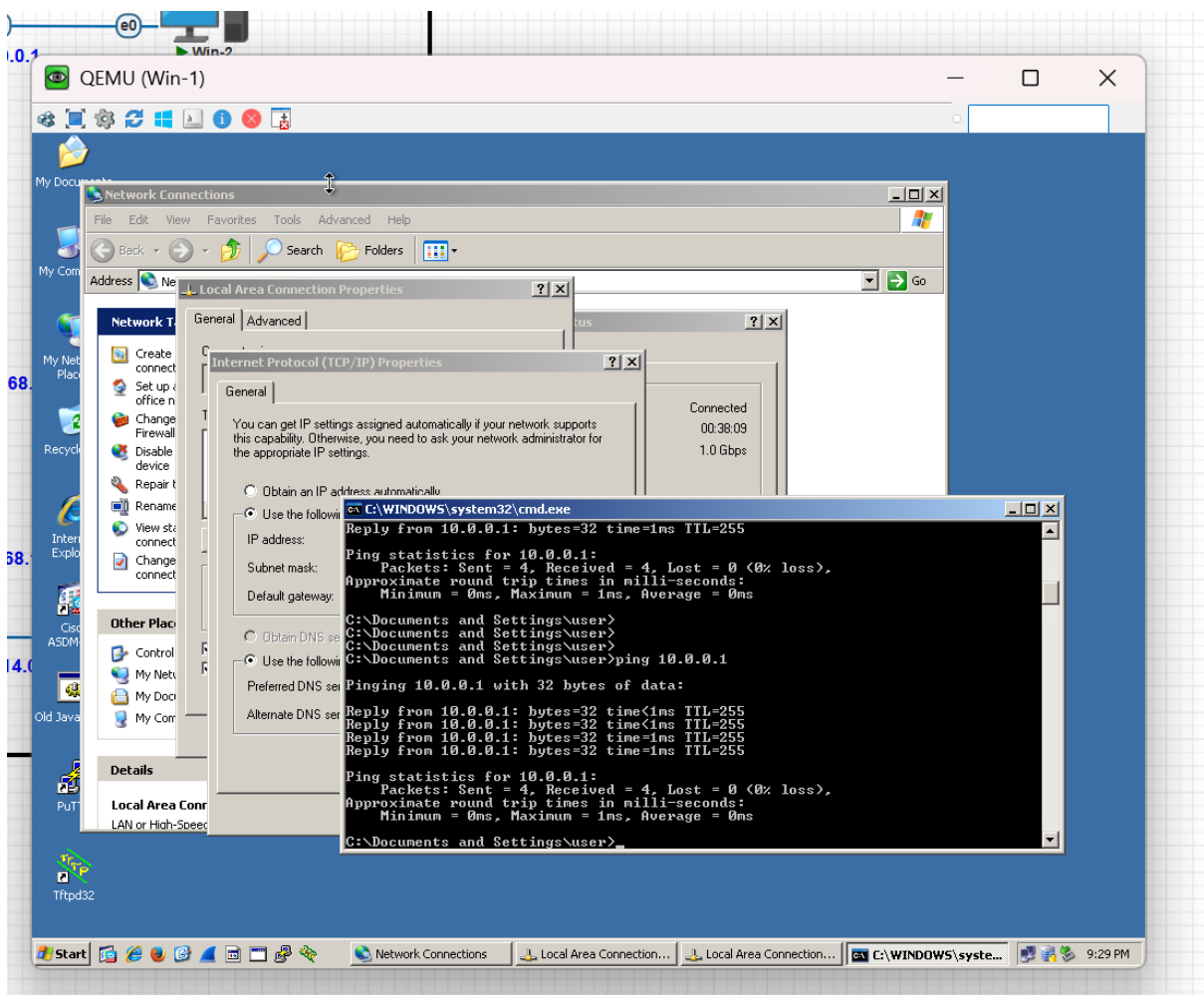
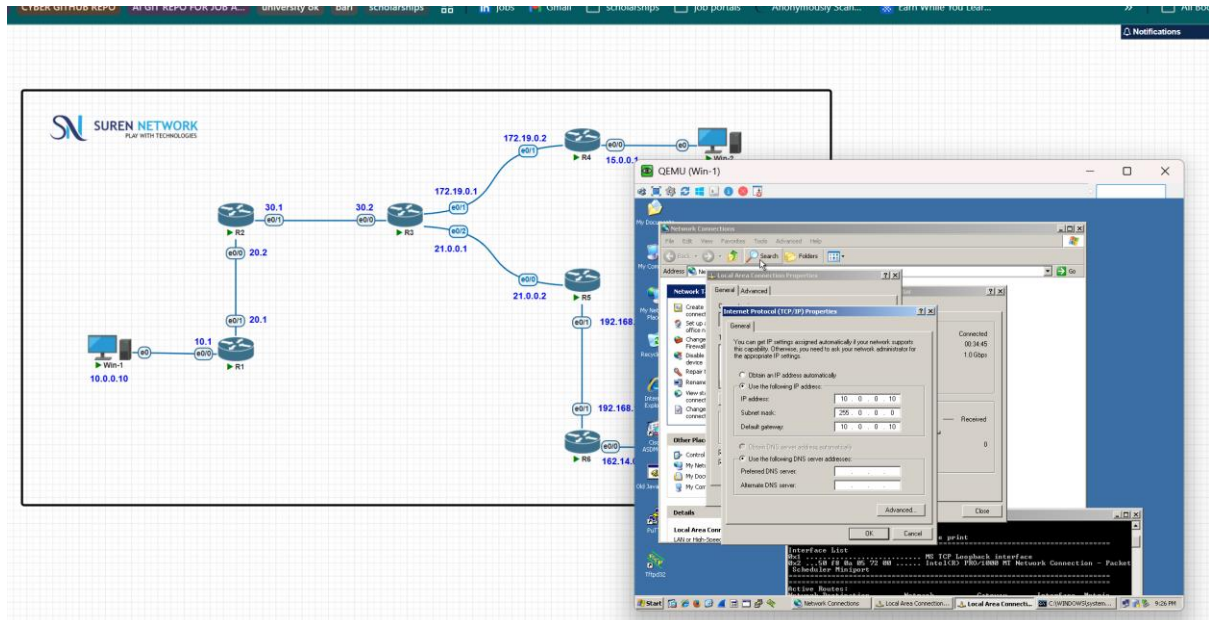
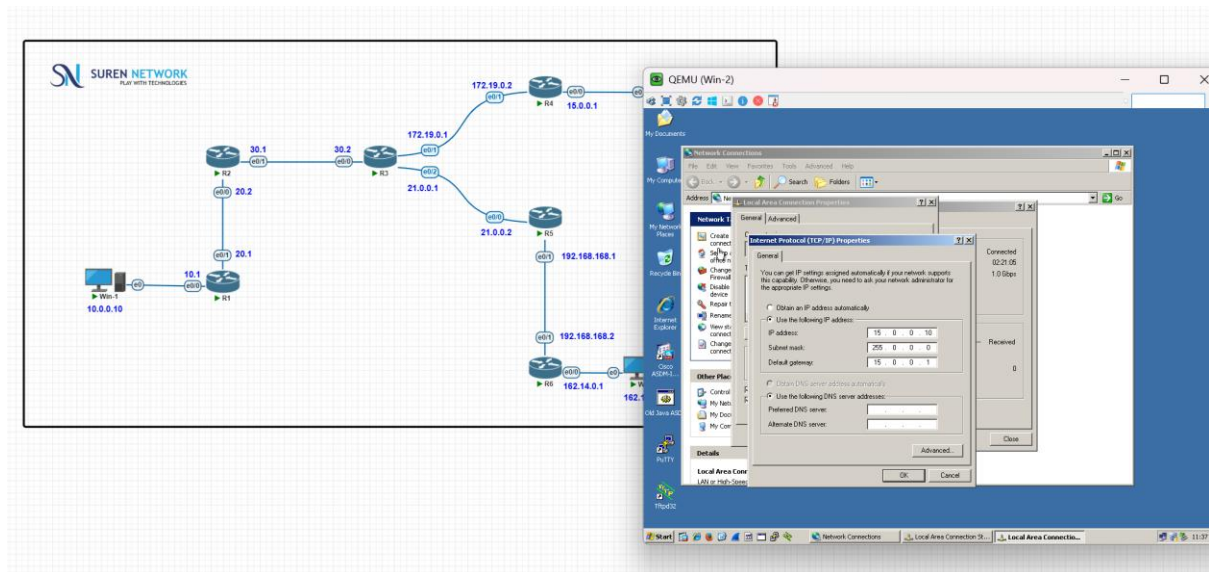


Project-1 static routing

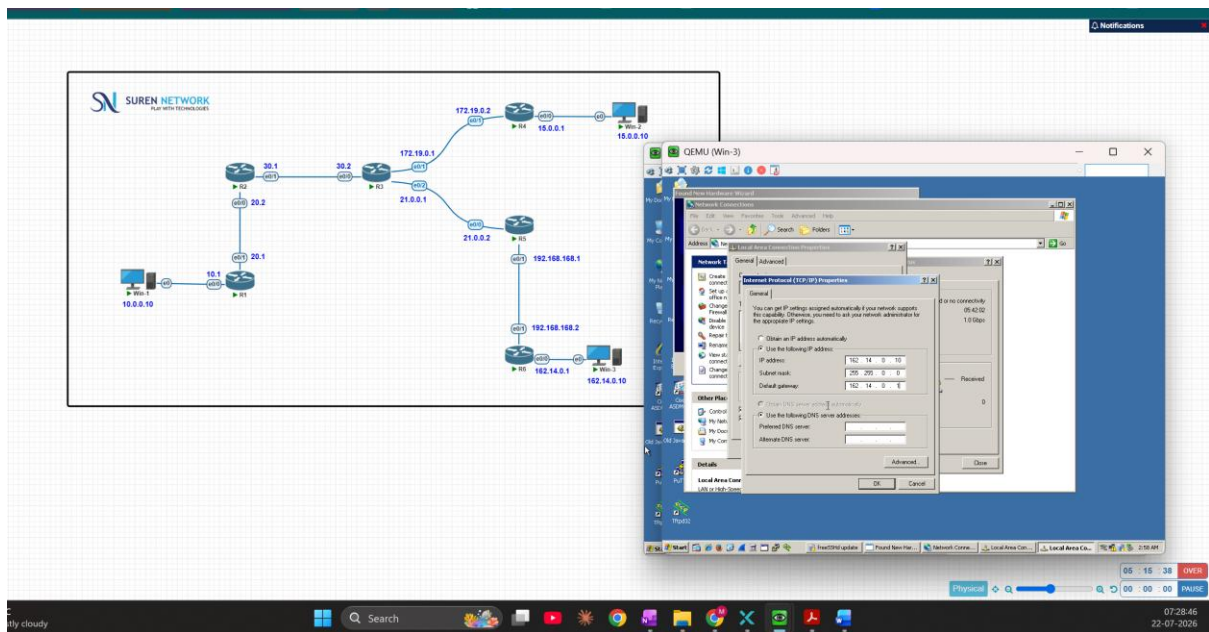
1.PC1: with of 10.0.0.10



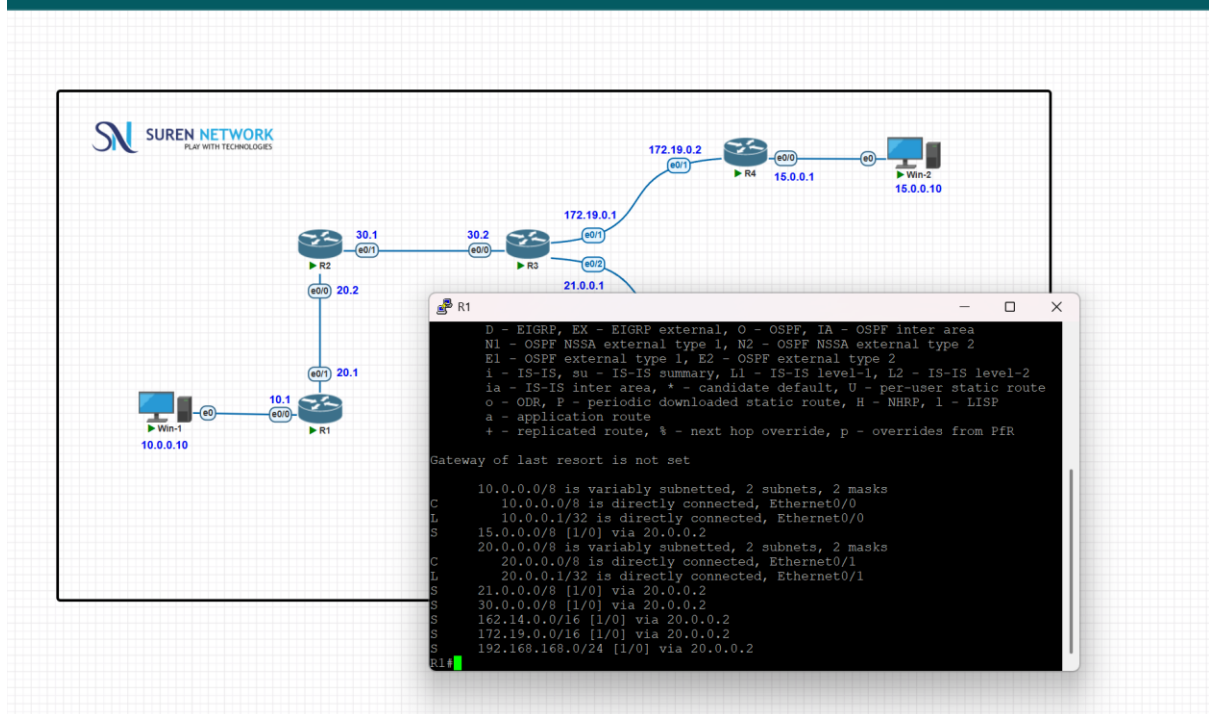
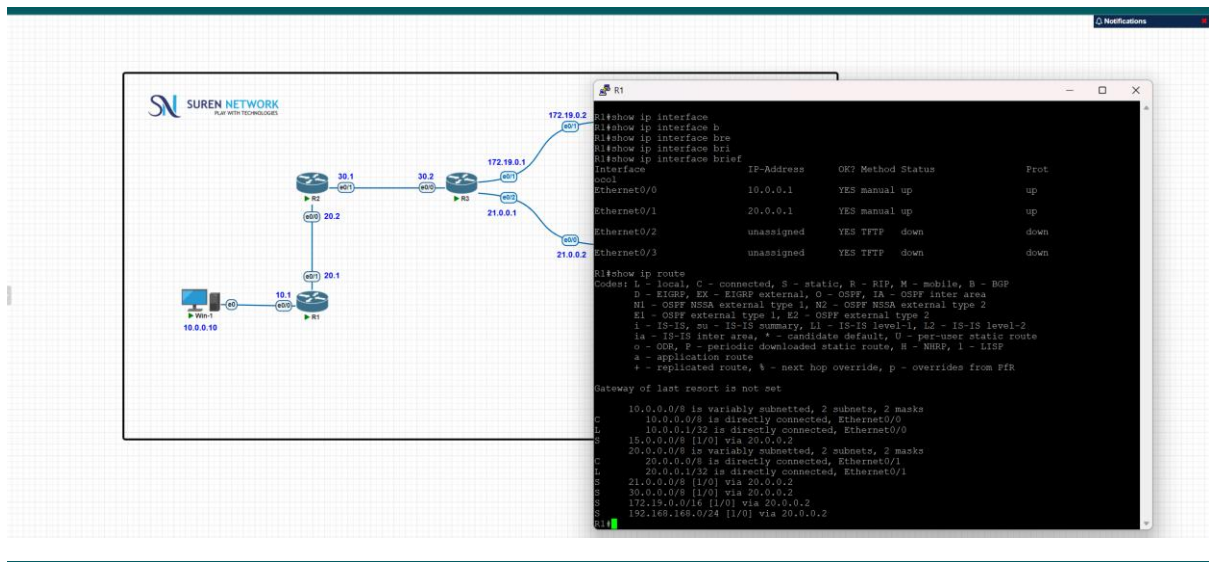
PC2: with of 15.0.0.10

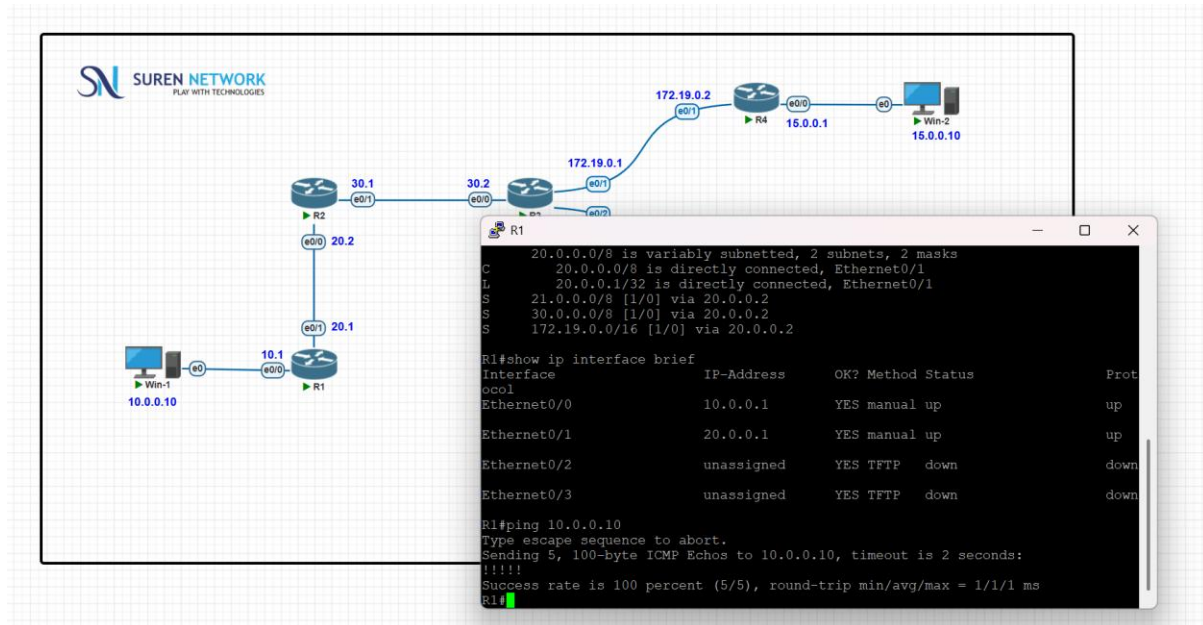


PC3: with of 162.14.0.10

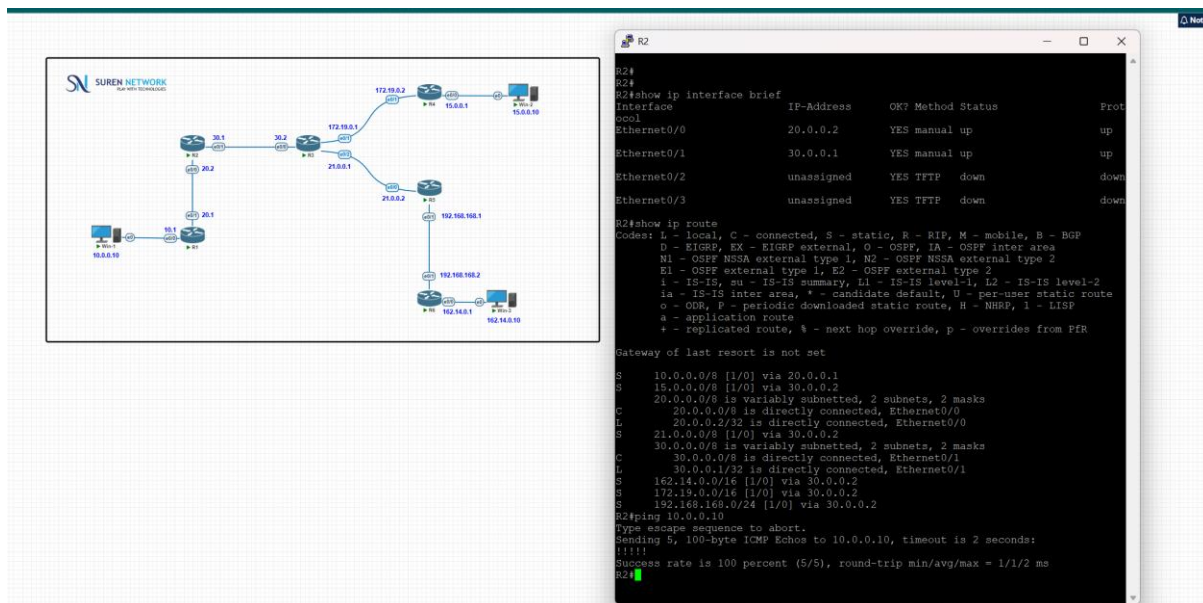


2.router1 – Ip and static route added





3. router2: Ip and static route added



4. router3: Ip and static route added

Router R3 Configuration:

```

R3#show ip int brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    30.0.0.2        YES manual up          up
Ethernet0/1    172.19.0.1      YES manual up          up
Ethernet0/2    21.0.0.1        YES manual up          up
Ethernet0/3    unassigned      YES TFTP  down        down

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, F - periodic downloaded static route, H - NHRP, I - ISIS
       a - application route
       * - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
S    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    21.0.0.0/8 is directly connected, Ethernet0/2
L    21.0.0.1/32 is directly connected, Ethernet0/2
S    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/0
C    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
S    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
R3#ping 10.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R3#ping 15.0.0.1

```

5. router4: Ip and static route added

Router R4 Configuration:

```

R4#show ip int brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    15.0.0.1        YES manual up          up
Ethernet0/1    172.19.0.2      YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

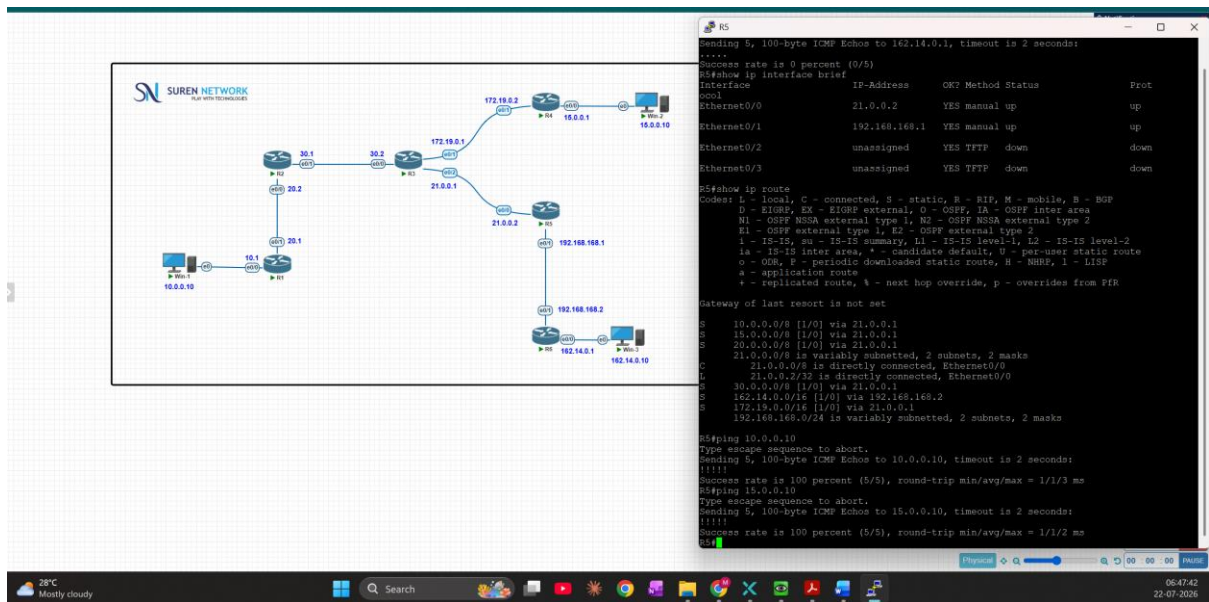
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, F - periodic downloaded static route, H - NHRP, I - ISIS
       a - application route
       * - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
S    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
S    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
R4#ping 10.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R4#ping 10.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
R4#

```

6. router5: Ip and static route added



The network diagram shows a topology with several routers (R1, R2, R3, R4, R5) and hosts. Router5 is connected to R4 and R3. The terminal output shows the configuration for Router5, including IP addresses and static routes.

```

R5#show ip interface brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    21.0.0.2        YES manual up          up
Ethernet0/1    192.168.168.1   YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down

R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

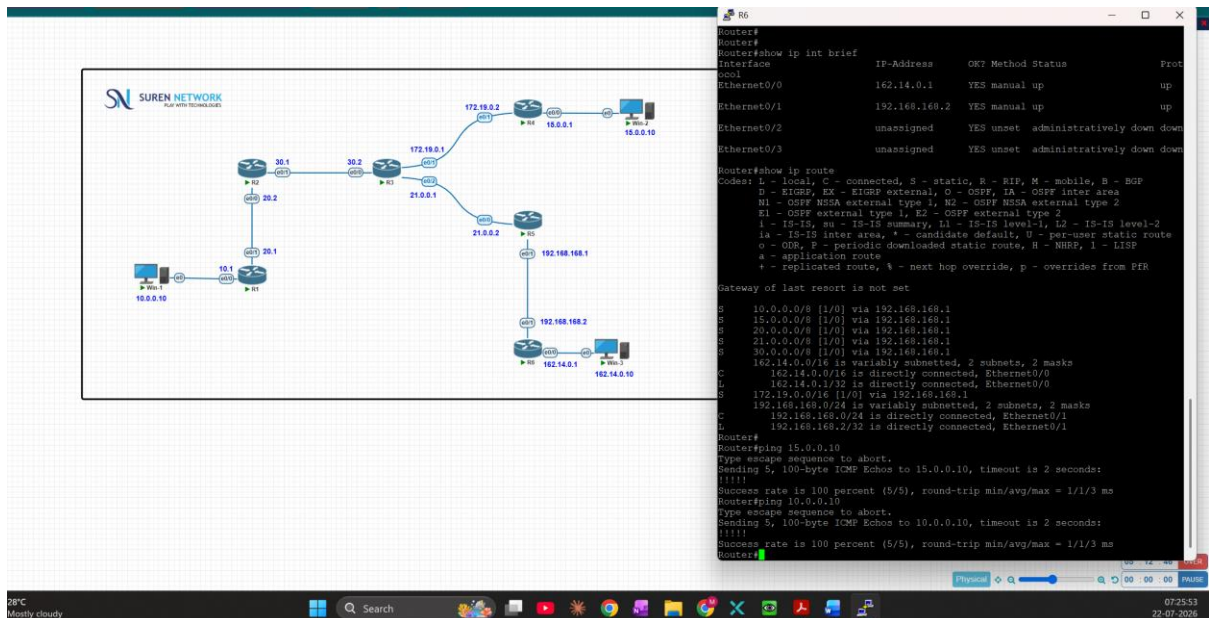
Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
S    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
       21.0.0.0/8 is directly connected, Ethernet0/0
       21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
S    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks

R5#ping 10.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
R5#ping 15.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 15.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R5#

```

7. router6: Ip and static route added



The network diagram shows the same topology as before, but with Router6 added. The terminal output shows the configuration for Router6, including IP addresses and static routes.

```

Router6#
Router6#show ip int brief
Interface      IP-Address      OK? Method Status      Prot
Ethernet0/0    162.14.0.1      YES manual up          up
Ethernet0/1    192.168.168.2   YES manual up          up
Ethernet0/2    unassigned      YES unset administratively down down
Ethernet0/3    unassigned      YES unset administratively down down

Router6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

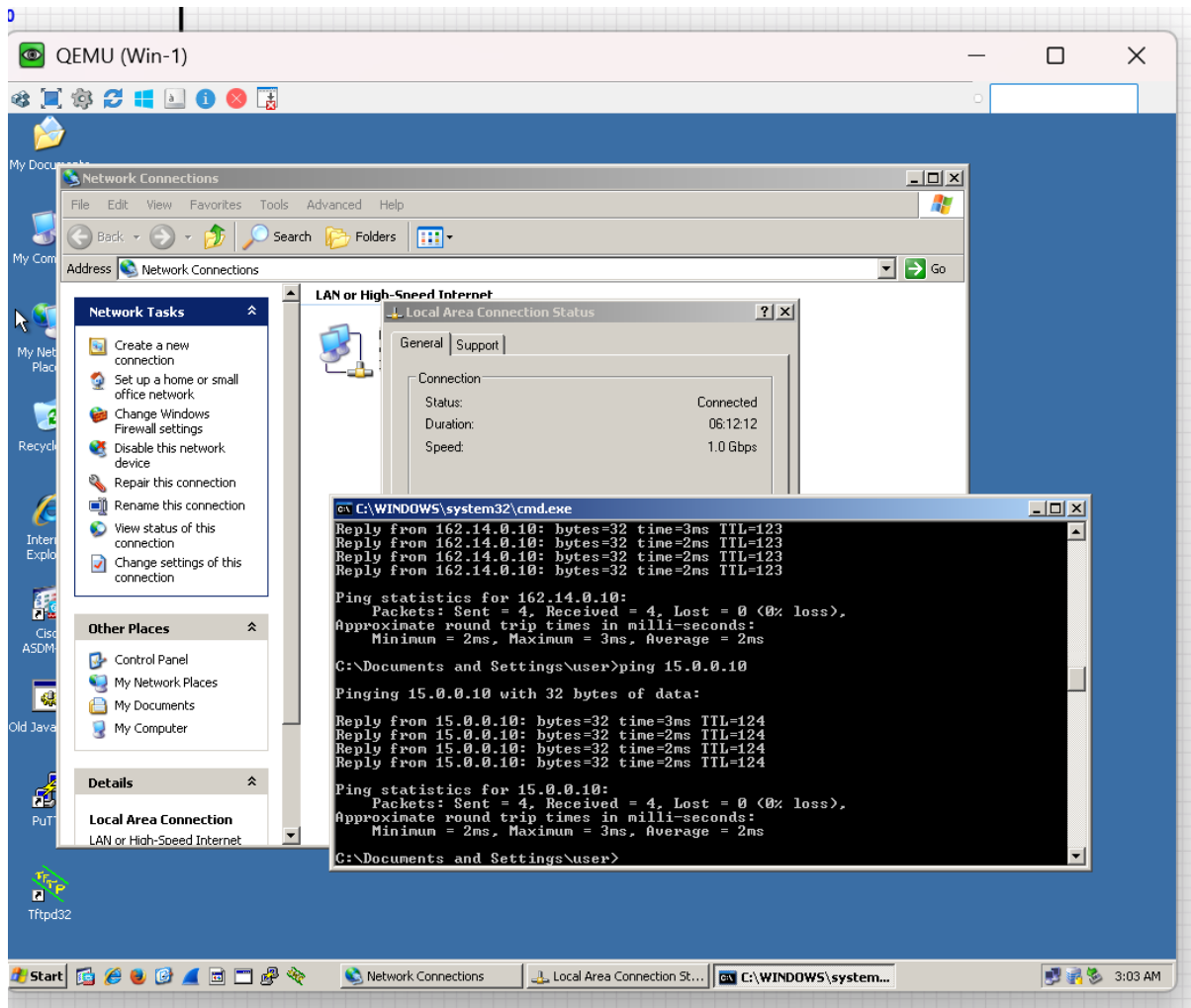
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
S    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
       162.14.0.0/16 is directly connected, Ethernet0/0
       162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
S    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
       192.168.168.0/24 is directly connected, Ethernet0/1
       192.168.168.2/32 is directly connected, Ethernet0/1

Router6#
Router6#ping 15.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 15.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router6#ping 10.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router6#

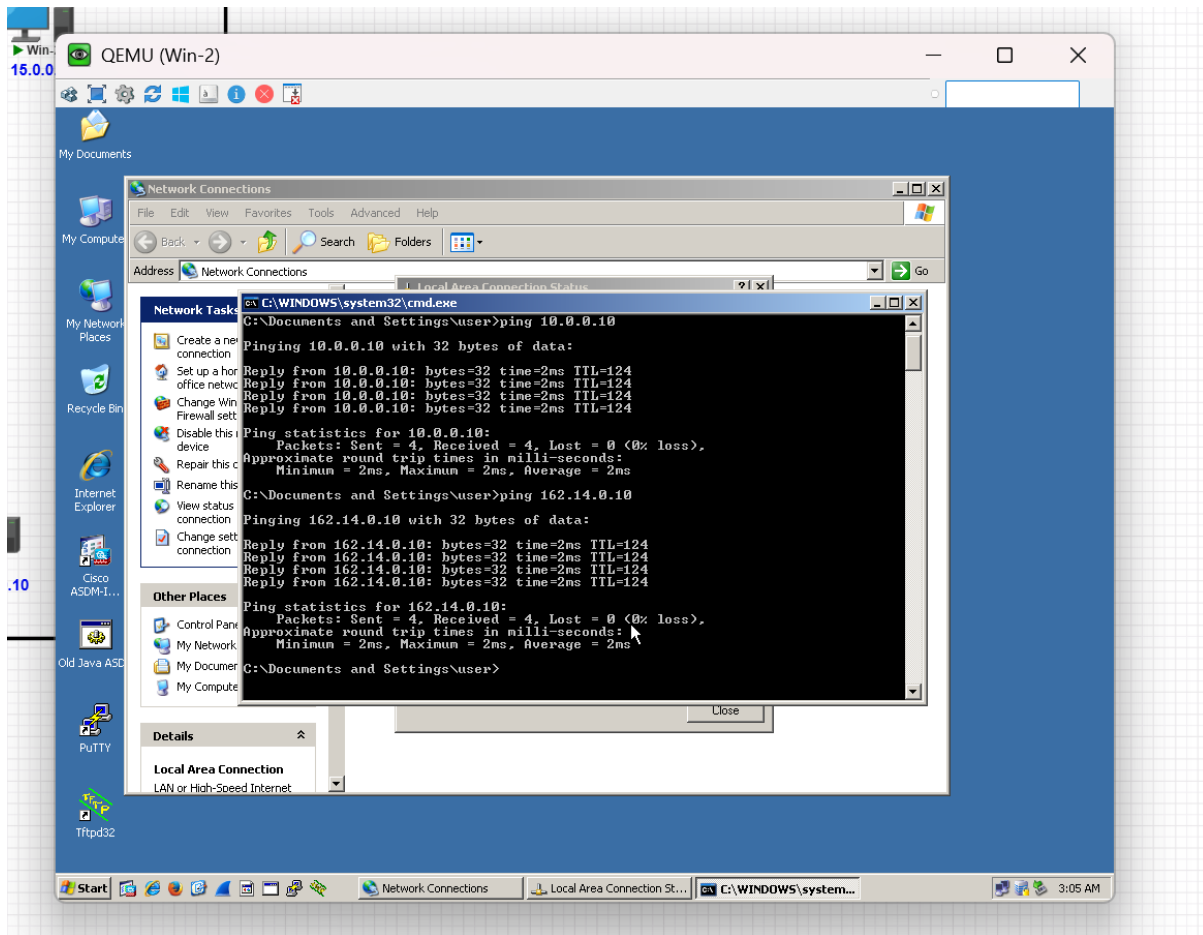
```

Connectivity check:

From 10.0.0.10 > Ping 15.0.0.10 and 162.14.0.10



From 15.0.0.10 > Ping 10.0.0.10 and 162.14.0.10



From 162.14.0.10 > Ping 10.0.0.10 and 15.0.0.10

